

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	XXXXXXX
Project Name	Smart-Lender-Loan-Eligibility-Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module/ Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter loan applicant details such as income, education, and credit history.	Frontend (HTML/CSS)	Done	High
2	Input Validation	Validates user input before processing.	Flask Backend	Done	High
3	Data Preprocessing	Cleans and transforms input data into the required format for prediction.	Preprocessing Module	Done	High
4	Loan Eligibility Prediction	Predicts whether the loan should be approved or rejected using the trained ML model.	Machine Learning Model	Done	High
5	Result Display	Displays the prediction result to the user.	Flask Application	Done	Medium
6	Trained Model Loading	Loads the saved machine learning model (model.pkl) for prediction.	Model Loader	Done	Medium
7	Responsive User Interface	Provides a user-friendly interface accessible through a web browser.	HTML/CSS/Java Script	Done	Medium
8	Error Handling	Handles invalid or incomplete user input gracefully.	Flask Backend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (User Input, Input Validation, Data Preprocessing, Loan Eligibility Prediction, Result Display)
Additional / Nice-to-Have Features	3 (Model Loading, Responsive UI, Error Handling)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	User Input Form, Result Display
2	Backend / Logic	3	Input Validation, Data Preprocessing, Loan Eligibility Prediction
3	Database / Storage	1	CSV Dataset, Trained Model (model.pkl)
4	API / Integration	1	Flask Application Routes
5	Security / Authentication	1	Input Validation and Basic Error Handling